

Question & Answer Text Statistics

Source: `Engdata(usage_binatanya_01-03-25_16-11-).csv`

Dataset: 95,515 turns across 51,571 conversations

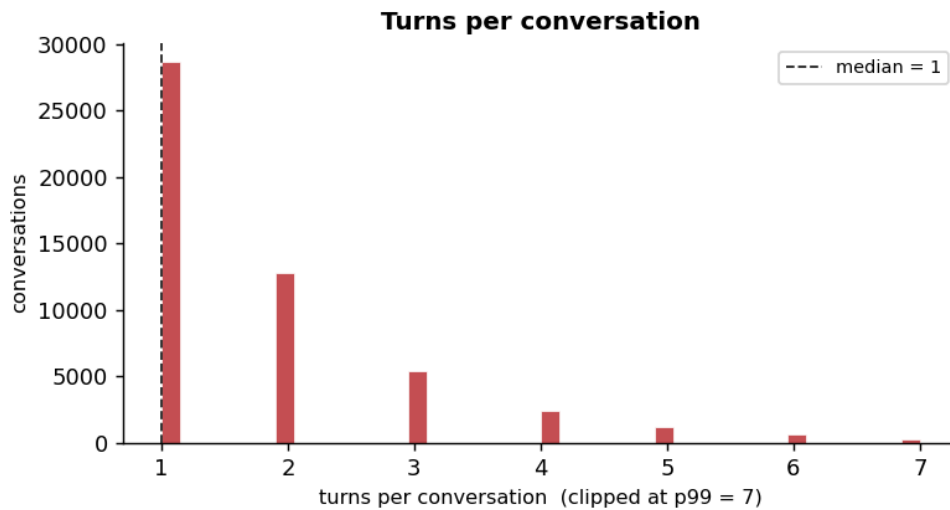
1. Coverage

Rows with empty English text are excluded from all downstream length and word statistics.

field	empty	usable
question (user_message_en)	679	94836
answer (agent_response_en)	197	95318

2. Conversation structure

Mean 1.85 turns per conversation, median 1, max 87. 28,632 conversations (56%) are single-turn.

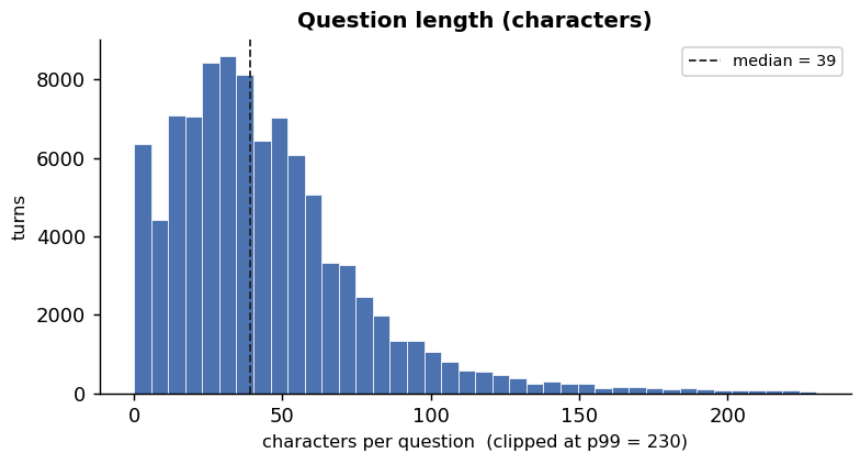
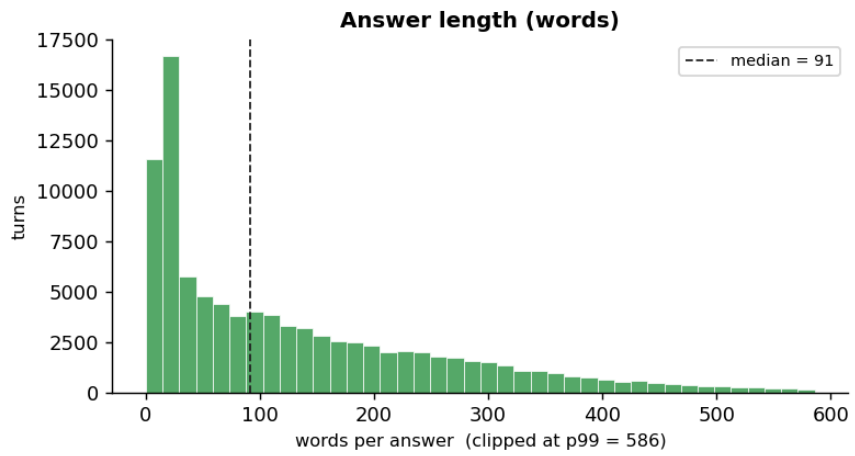
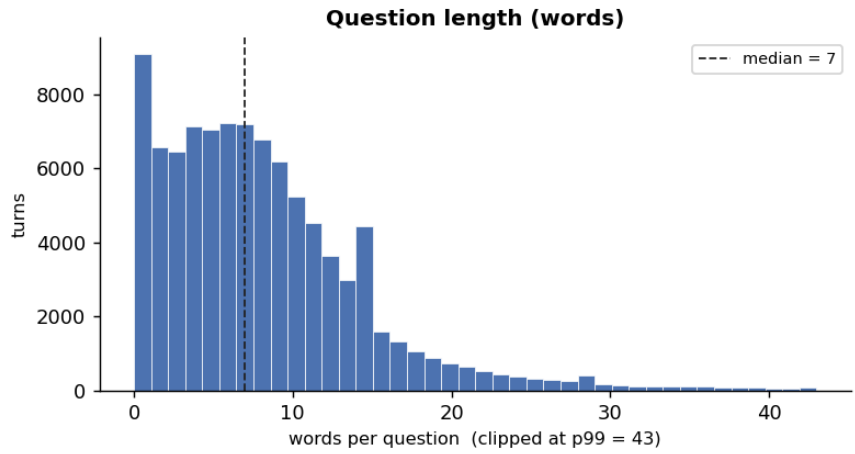


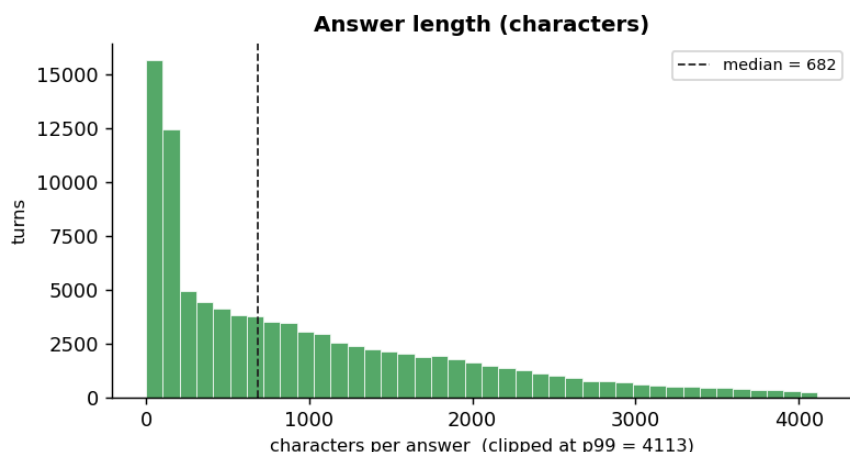
3. Length distributions

Questions are short; answers are ~15x longer. Percentiles (p25–p99) shown because both distributions are right-skewed by outliers.

metric	count	mean	std	min	max	p25	p50	p75	p90	p95	p99
question_chars	95515	48.2	67.4	0	11279	22.0	39.0	60.0	88.0	114.0	230.0
question_words	95515	8.8	11.7	0	1797	4.0	7.0	11.0	16.0	21.0	43.0
answer_chars	9551	1000.	1010.	0	9221	138.	682.	1550.	2474.	3105.	4112.

	5	9	3			0	0	0	0	0	9
answer_words	9551 5	135.8	137.6	0	1077	23.0	91.0	207.0	332.0	414.0	586.0





4. Feedback

Only 1,453 of 95,515 turns (1.5%) were rated. Of those: 761 thumbs-up, 692 thumbs-down — 52.4% satisfaction. The low rating rate means this signal is too sparse to segment on reliably.

5. Most frequent words

Questions

what (9,813), property (8,574), parking (7,782), want (6,728), tax (5,694), number (4,598), pay (4,576), report (4,234), need (4,123), street (3,779), know (3,543), like (3,019), payment (2,934), ticket (2,781), email (2,620), taxes (2,548), hello (2,419), who (2,399), form (2,386), municipality (2,229), school (2,204), received (2,111), beach (2,056), department (2,044), today (2,032)

Answers

parking (113,358), information (90,797), property (88,305), municipality (82,904), city (74,550), website (60,543), department (57,475), tax (54,257), payment (52,194), form (49,853), contact (48,382), online (41,355), help (40,952), municipal (37,207), report (36,707), residents (36,118), application (35,830), center (35,714), permit (34,659), request (34,646), between (34,416), metropark (34,366), street (32,942), discount (31,589), address (31,429)

6. Notes & data-quality flags

- Longest question is 11,279 chars / 1,797 words — almost certainly a paste/junk outlier; consider capping questions at p99 (~43 words).
- Answer length p99 is ~4,113 chars; a few answers reach 9,221 chars. Cap or flag before modeling.

- Histograms are clipped at p99 so the long right tail does not flatten the visible distribution; tables report the true min/max.

7. Combined Q+A topic model (LDA)

An additional LDA was fit on a merged document per conversation, where the question text (user_message_en) and the answer text (agent_response_en) are concatenated into one paragraph.

7.1 Combined topics (k = 16)

topic	top keywords
0	parking, report, metropark, select, website, appeal, payment, pay, ticket, request, department, action, online, search, authorityid
1	street, city, information, shelter, address, located, neighborhood, removal, shelters, kiryat, help, public, herzl, waste, evacuation
2	building, information, construction, engineering, application, business, cityhall, planning, permit, complex, plan, municipality, renewal, licensing, city
3	discount, property, square, tax, application, discounts, apartment, meters, income, senior, area, insurance, national, disabled, residential
4	parking, permit, city, streets, residents, free, resident, vehicle, discount, white, blue, select, application, provides, area
5	property, tax, payment, municipality, pay, payments, online, form, credit, taxes, bank, standing, account, direct, order
6	park, garden, city, public, information, gardens, id, facilities, municipality, located, lighting, street, areas, publicgardens, area
7	tender, tenders, municipality, education, submission, deadline, kindergarten, kindergartens, external, gan, information, details, municipal, date, special
8	department, contact, phone, information, center, municipality, sunday, thursday, address, service, street, reception, pm, hours, floor
9	information, municipal, services, help, water, city, traffic, municipality, street, provide, public, work, sorry, regarding, hello
10	beach, independence, beaches, pm, city, events, information, day, sunset, hour, sironit, half, sanz, bathing, services
11	contact, municipality, hotline, municipal, request, form, website, service, number, card, information, waste, resident, understand, department
12	information, center, residents, centers, activities, senior, citizens, club, municipality, social, offers, services, courses, city, welfare
13	pm, city, events, details, nis, class, tickets, purchase, place, held, classes, center, community, cost, information
14	property, form, forms, process, tax, cityforms, code, tat, sug, nose, processoneclick, change, taboo,

	appbuilder, certificate
15	registration, education, school, kindergarten, parents, student, form, placement, residents, documents, transfer, request, special, address, website

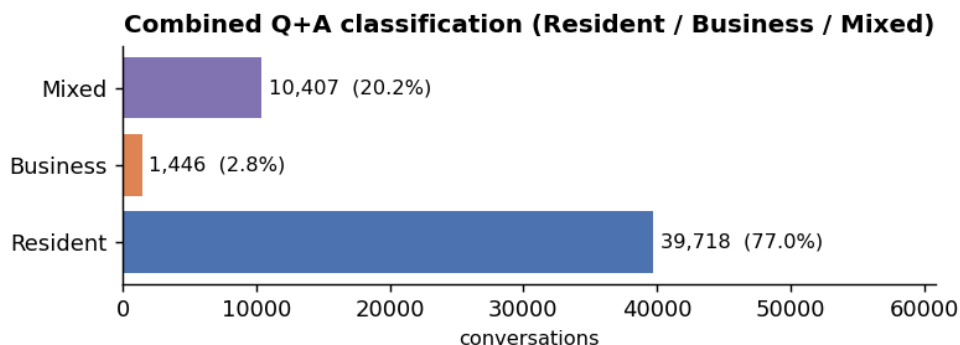
7.2 Topic-to-category map

Mixed topics are ones whose keywords legitimately serve both residents and businesses (building permits, generic department contact, generic municipal services), so a Mixed label means the conversation is genuinely ambiguous rather than misclassified.

category	topic	name
Resident	0	Parking ticket / appeal
Resident	1	Street, shelter, waste
Resident	3	Property tax discount (senior/disabled)
Resident	4	Parking permit (resident)
Resident	5	Property tax payment
Resident	6	Parks & gardens
Resident	10	Beach & events
Resident	11	Hotline & resident services
Resident	12	Senior citizens & welfare
Resident	13	Events & community classes
Resident	14	Property tax forms
Resident	15	School & kindergarten registration
Business	7	Tenders & procurement
Mixed	2	Building permit + business licensing
Mixed	8	Contact / department info
Mixed	9	General municipal services

7.3 Label distribution

label	conversations	share
Resident	39,718	77.0%
Business	1,446	2.8%
Mixed	10,407	20.2%



8. Per-side classification: question vs answer (separate LDA models)

This section uses the separate question-side and answer-side LDA models. Each conversation's dominant topic on each side is mapped to Resident / Business / Neutral, and three labelling modes are reported: from the question alone, from the answer alone, and from both sides agreeing.

11.1 Topic-to-sector map

Question-side (user_topic_*, k = 4):

topic	sector	top keywords
user_topic_0	Resident	street, shelter, area, near, parking, building, construction, neighborhood, lot, removal, kiryat, live, waste, city, garbage
user_topic_1	Mixed	email, need, school, number, department, phone, kindergarten, address, want, municipality, permit, hi, contact, looking, form
user_topic_2	Resident	beach, today, open, day, independence, hi, does, events, hours, time, know, city, square, tomorrow, performances
user_topic_3	Resident	property, parking, tax, pay, want, report, payment, ticket, taxes, number, received, hi, discount, need, check

Answer-side (agent_topic_*, k = 15):

topic	sector	top keywords
agent_topic_0	Resident	education, registration, kindergarten, school, parents, student, residents, special, kindergartens, placement, address, information, transfer, children, documents
agent_topic_1	Business	tender, municipality, tenders, submission, deadline, social, welfare, external, education, details, administration, department, municipal, services, date
agent_topic_2	Resident	discount, property, square, tax, discounts, meters, apartment, application, income, senior, area, national, insurance, residential, disabled
agent_topic_3	Mixed	building, information, city, shelter, street, permit, engineering, plan, shelters, planning, municipality, public, cityhall, construction, plans
agent_topic_4	Resident	property, tax, form, forms, process, cityforms, code, tat, sug, nose, processoneclick, change,

		signed, appbuilder, taboo
agent_topic_5	Resident	center, information, school, class, pm, community, city, classes, residents, centers, nis, register, street, phone, club
agent_topic_6	Mixed	property, payment, tax, municipality, payments, online, credit, pay, form, bank, direct, standing, order, card, debit
agent_topic_7	Resident	parking, permit, city, resident, streets, residents, vehicle, free, discount, card, white, select, application, blue, provides
agent_topic_8	Business	form, request, application, building, documents, municipality, committee, exemption, property, pdf, department, submitted, days, support, submit
agent_topic_9	Resident	city, removal, waste, information, street, address, help, work, taken, ben, municipality, evacuation, database, carried, traffic
agent_topic_10	Business	business, license, licensing, information, dog, signage, application, law, municipality, event, noise, fee, environmental, appointment, veterinary
agent_topic_11	Resident	beach, information, complex, city, public, park, beaches, garden, renewal, area, gardens, urban, street, located, accessible
agent_topic_12	Mixed	contact, municipality, department, information, municipal, phone, help, services, hotline, service, website, thursday, public, sunday, regarding
agent_topic_13	Resident	parking, report, metropark, select, website, appeal, payment, request, department, pay, ticket, action, online, search, authorityid
agent_topic_14	Resident	pm, events, city, independence, details, tickets, purchase, place, square, free, information, day, nis, admission, event

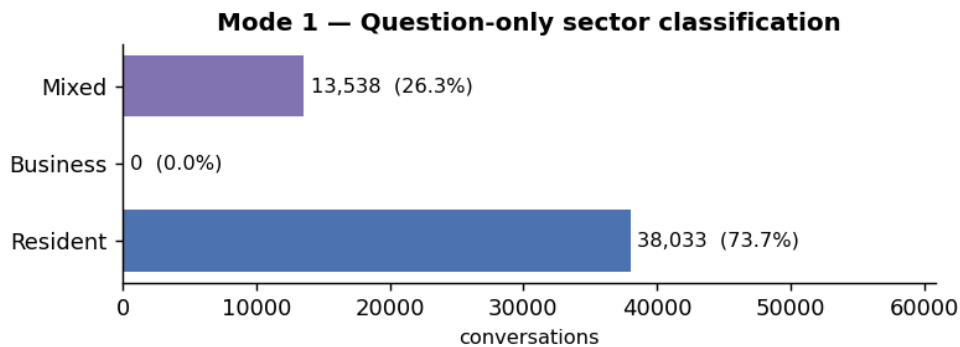
- Note: no user topic is mapped to Business — Business intent shows up in user_topic_1 (Mixed) and in the answer side. This means Mode 3 (both sides agree) can never produce a Business label; conversations whose answer side is Business fall into Uncategorized in Mode 3.

11.2 Mode 1 — Question only

Sector taken from the dominant user topic. The Business count is zero by construction (see map).

label	conversations	share
Resident	38,033	73.7%
Business	0	0.0%

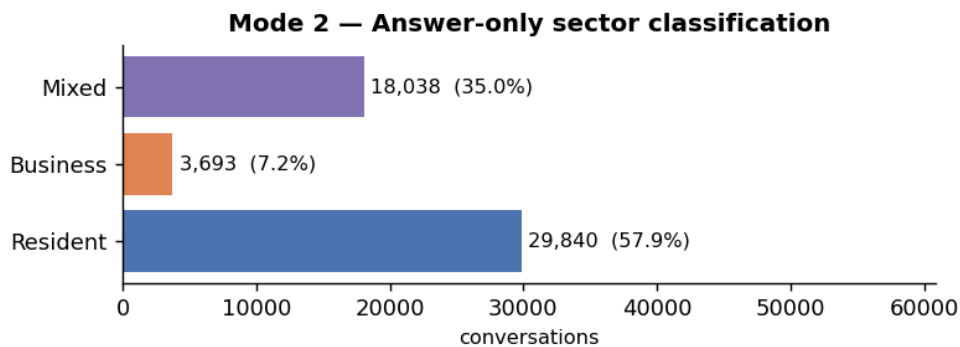
Mixed	13,538	26.3%
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11.3 Mode 2 — Answer only

Sector taken from the dominant agent topic. All three sectors are reachable here.

label	conversations	share
Resident	29,840	57.9%
Business	3,693	7.2%
Mixed	18,038	35.0%

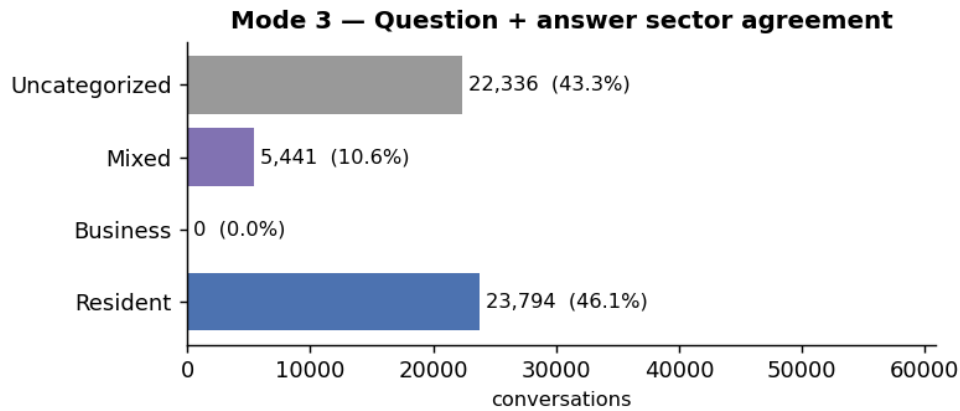


11.4 Mode 3 — Both sides must agree

A sector is assigned only when the question-side sector and the answer-side sector are the same Resident/Business/Mixed value; otherwise the conversation is Uncategorized.

Agreement rate (question and answer give the same sector): $29,235 / 51,571 = 56.7\%$.

label	conversations	share
Resident	23,794	46.1%
Business	0	0.0%
Mixed	5,441	10.6%
Uncategorized	22,336	43.3%



11.5 Question × Answer cross-tab

Rows = question-side sector, columns = answer-side sector. The diagonal cells are the conversations Mode 3 keeps; off-diagonal cells become Uncategorized.

question \ answer	Resident	Business	Mixed
Resident	23,794	1,642	12,597
Business	0	0	0
Mixed	6,046	2,051	5,441